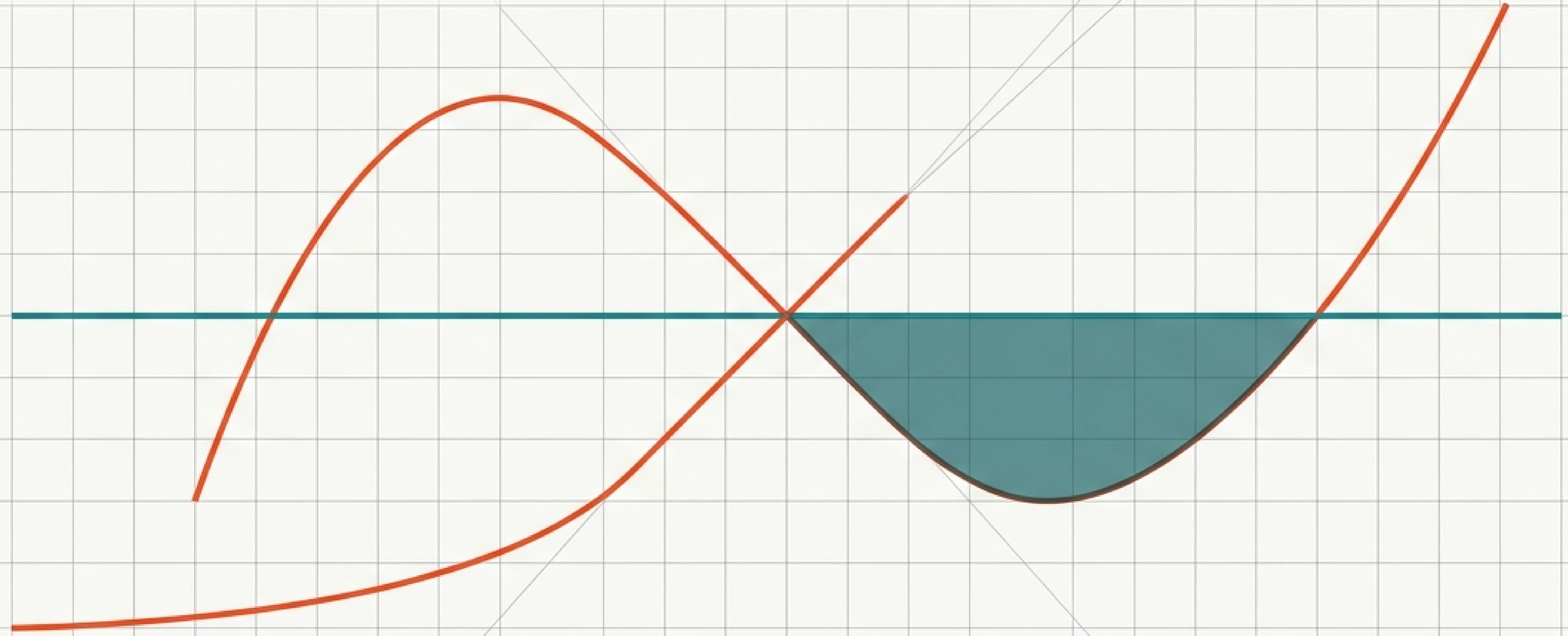
The Threshold of Deferral

A Crossover Theorem for Maintenance and Repair



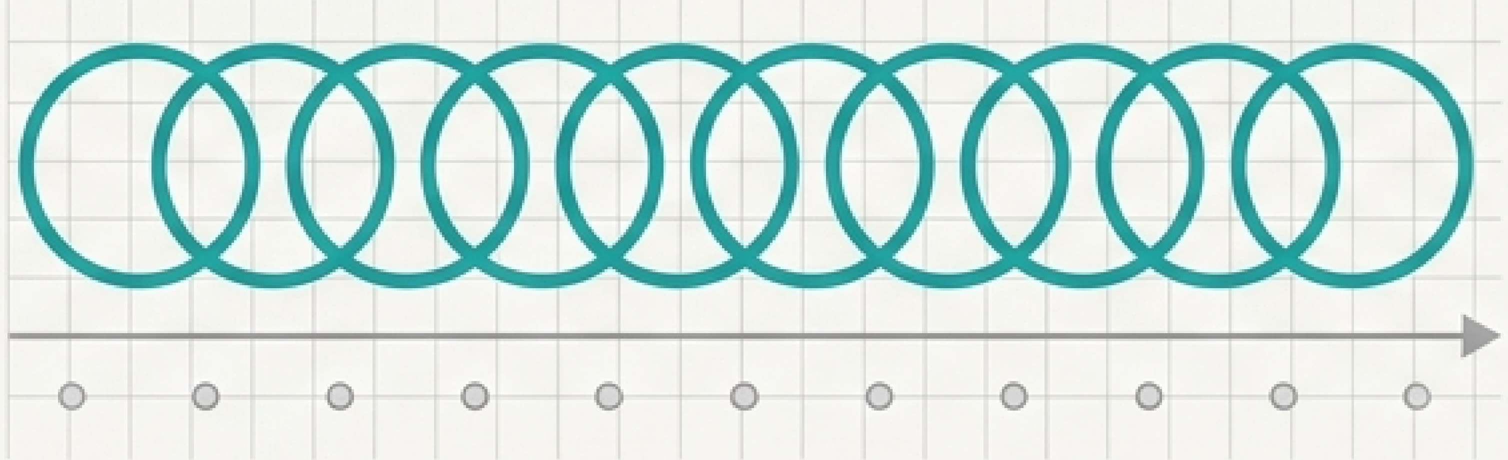
Flyxion, 2026

An executive briefing on the mathematical regimes of system correction.

Two distinct kinds of correction

Maintenance (M_t)

Unconditional



Acts on every step to reduce deviation before it compounds.

Repair (ρ_t)

Conditional

Triggered only when the system state (x_t) leaves the viability manifold.

The core question is not whether maintenance is preferable, but under what exact conditions a rational system switches from one to the other.

The dynamics of uncorrected deviation.

Equation Anatomy

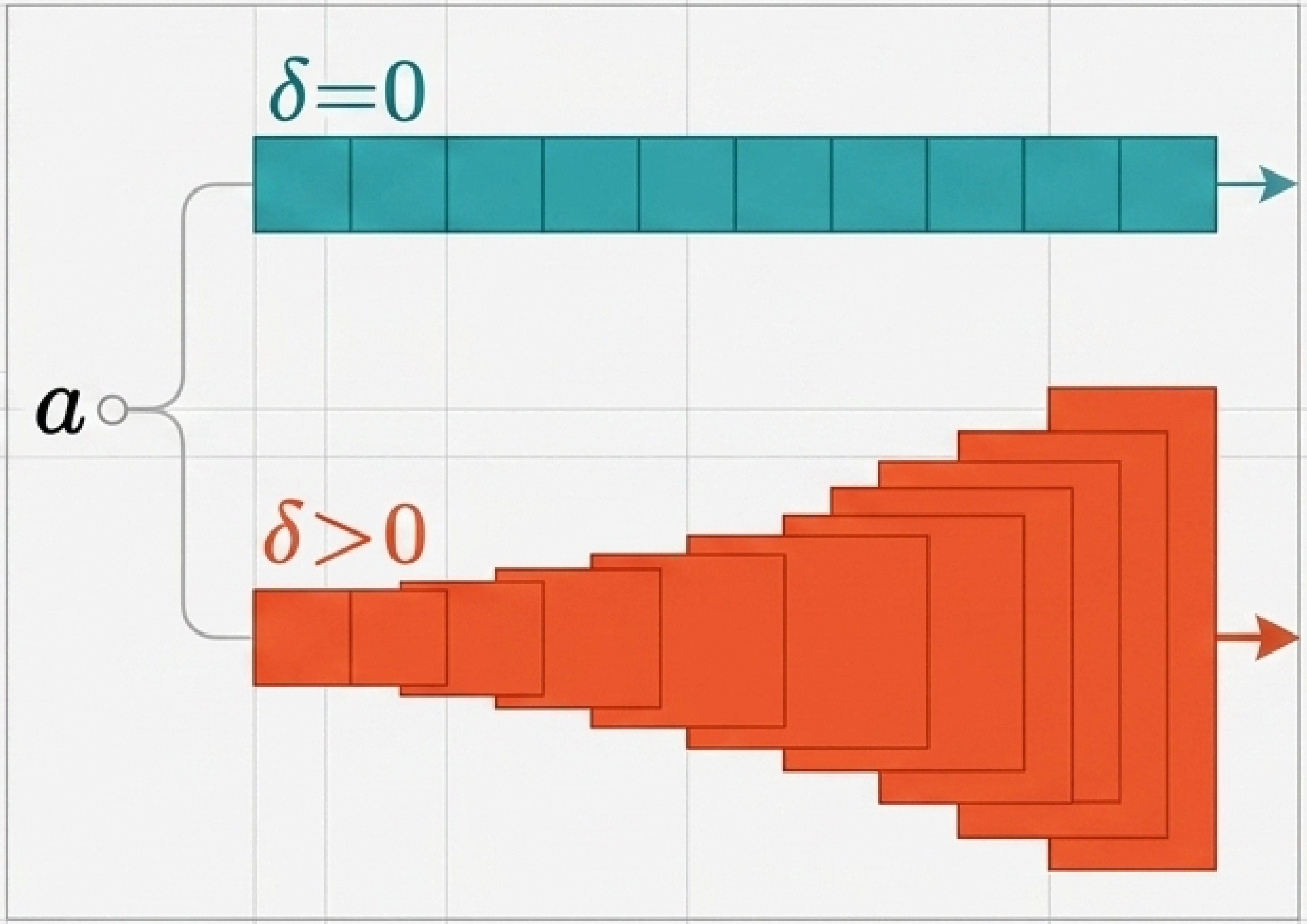
a: Deviation introduced at each step.

$$d_{t+1} = (1 + \delta)(d_t - m_t) + a$$

m_t: Maintenance applied before compounding.

δ: The delay-amplification rate.

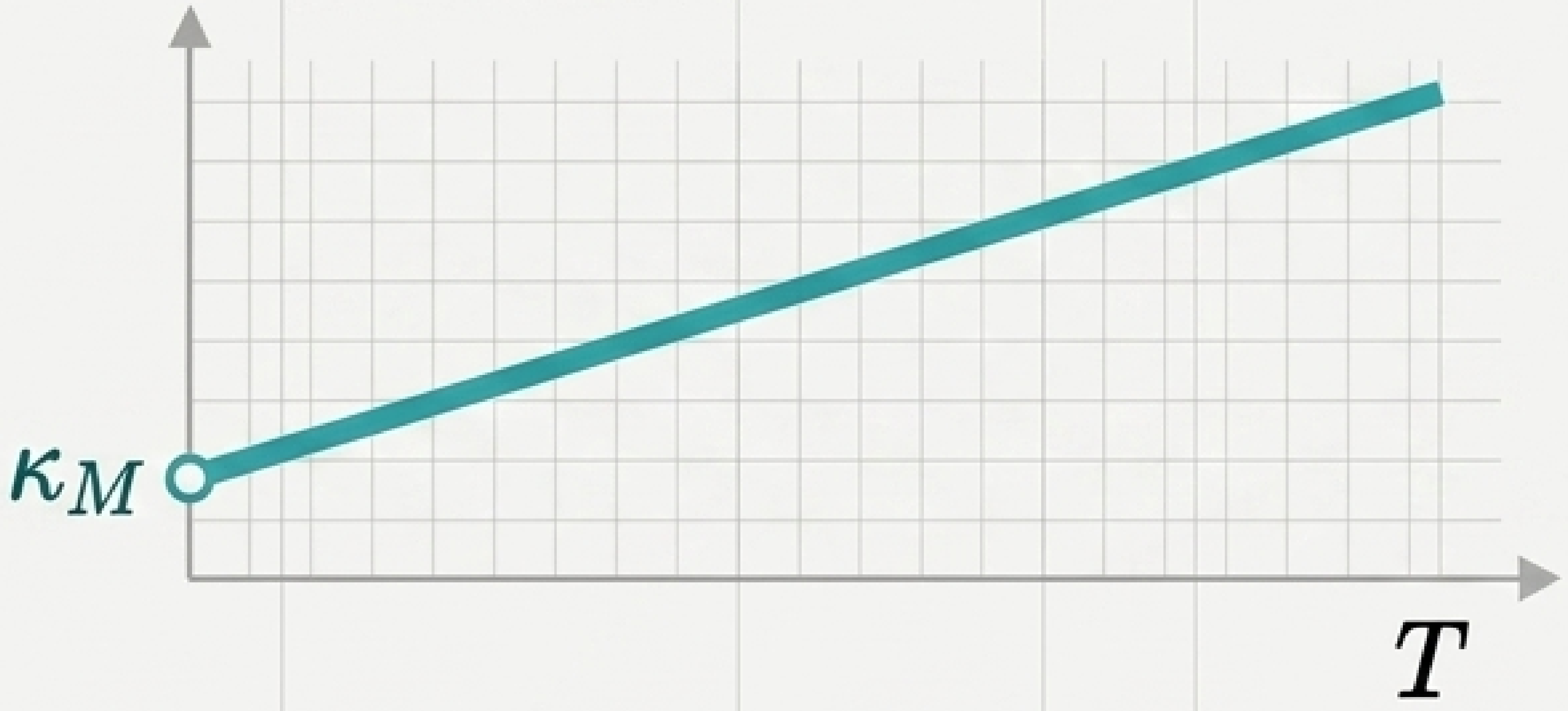
Branching Growth Visual



$\delta = 0$ is a distinct regime of pure accumulation without active amplification (e.g., dormant defects).

The competing cost functionals.

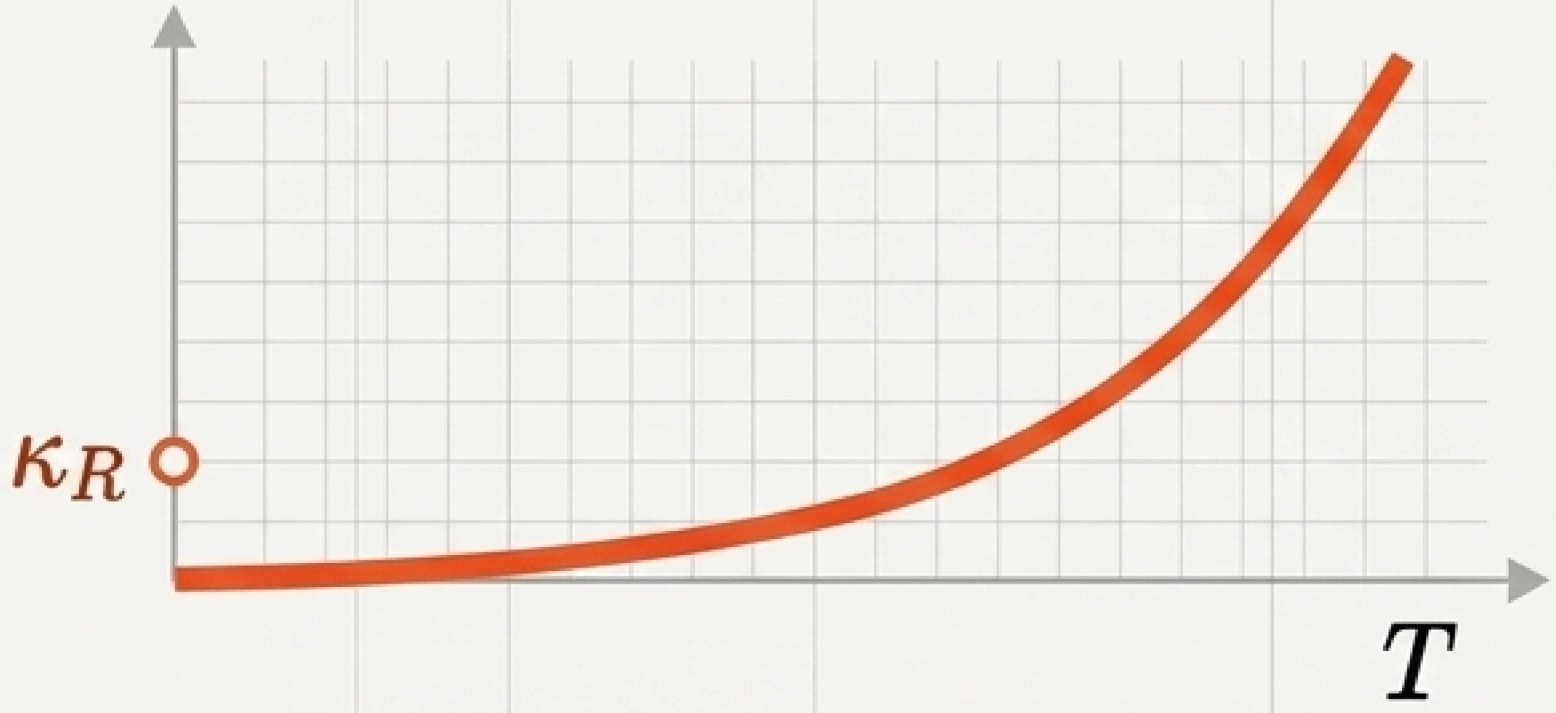
Maintenance Cost (C_M)



$$C_M(T) = T(\kappa_M + \alpha a^p)$$

Pays fixed maintenance overhead (κ_M) every step. Growth is linear.

Repair Cost (C_R)



$$C_R(T) = \kappa_R + \alpha(a/\delta)^p((1+\delta)^T - 1)^p$$

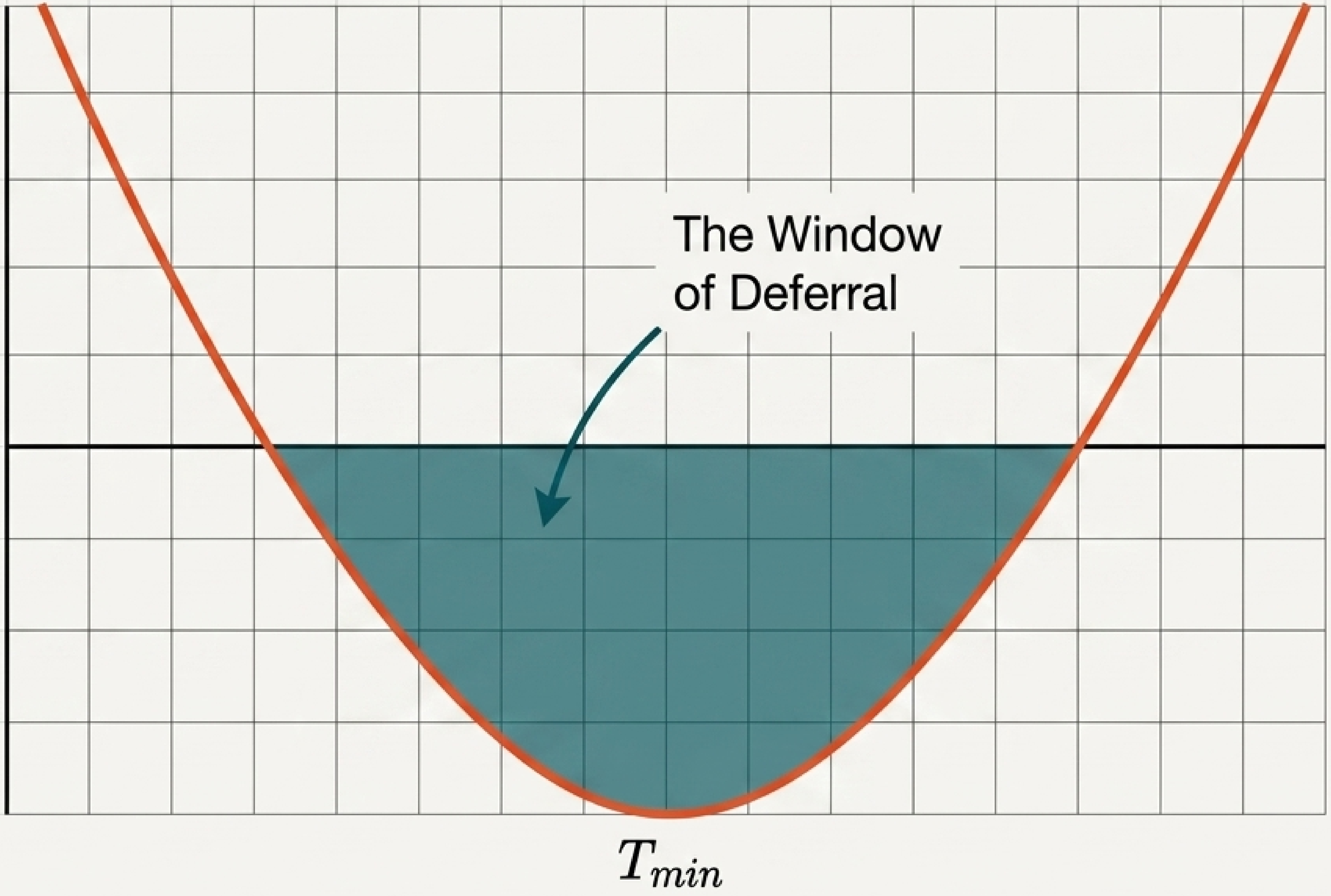
Pays fixed repair overhead (κ_R) once. Growth is convex ($p > 1$).

C_R always defers correction to a single convex-cost operation at the end of the horizon T.

The convex differential.

$$D(T) = C_R(T) - C_M(T)$$

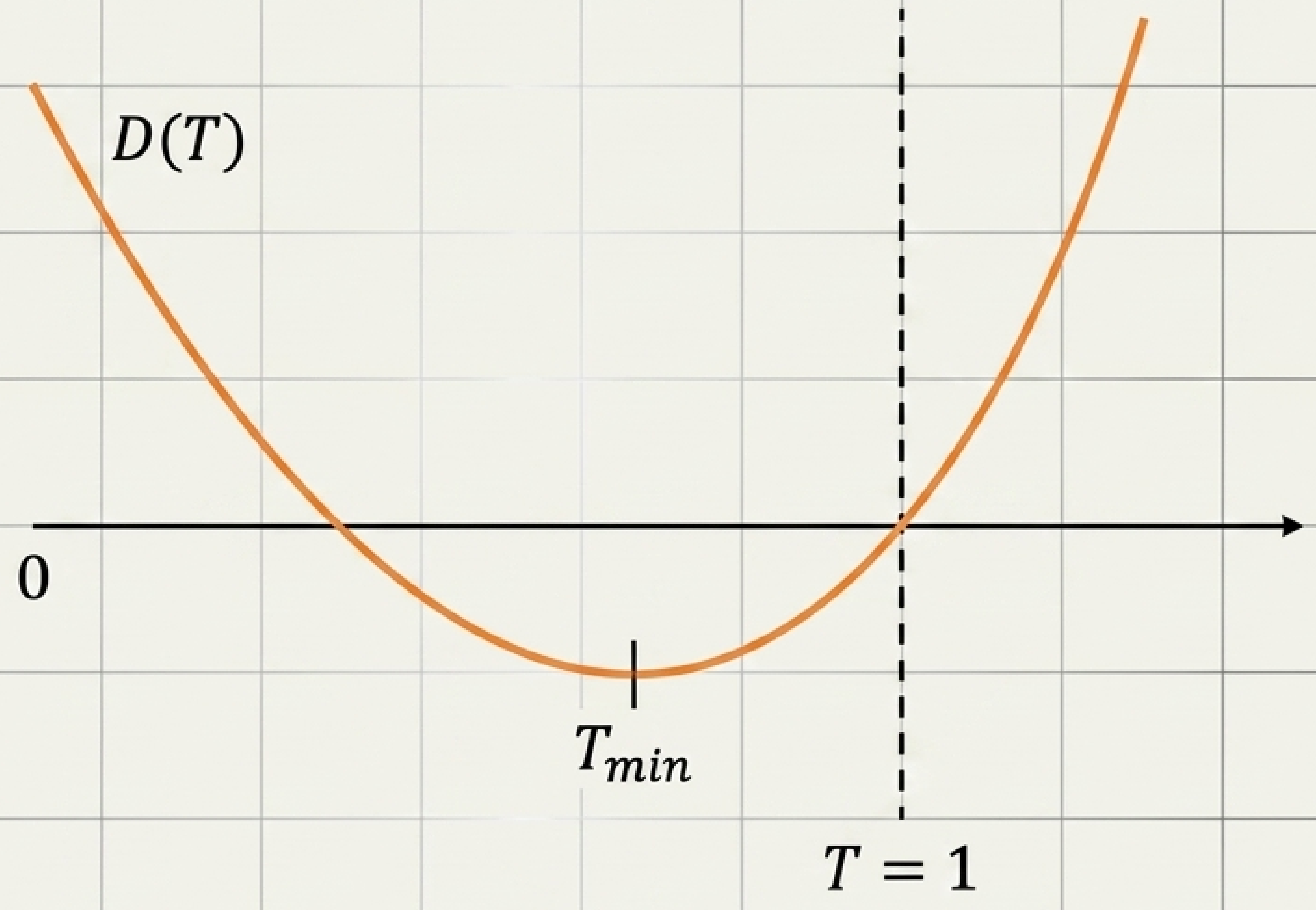
- Because C_R is convex and C_M is affine, their difference $D(T)$ is strictly convex.
- A convex function crossing a fixed level can have at most two roots. The roots straddle the unconstrained minimum T_{min}.
- Where $D(T) < 0$, deferred repair is the economically rational choice.



The Early Recovery Condition ($T_{min} \leq 1$)

- Threshold existence is decided in the region $T \geq 1$. If $T_{min} \leq 1$, the system has already passed the point of maximum relative advantage for deferred repair by the time the shortest possible horizon is reached.

- The curve is already on its ascending branch. Delaying further yields rapidly diminishing returns.

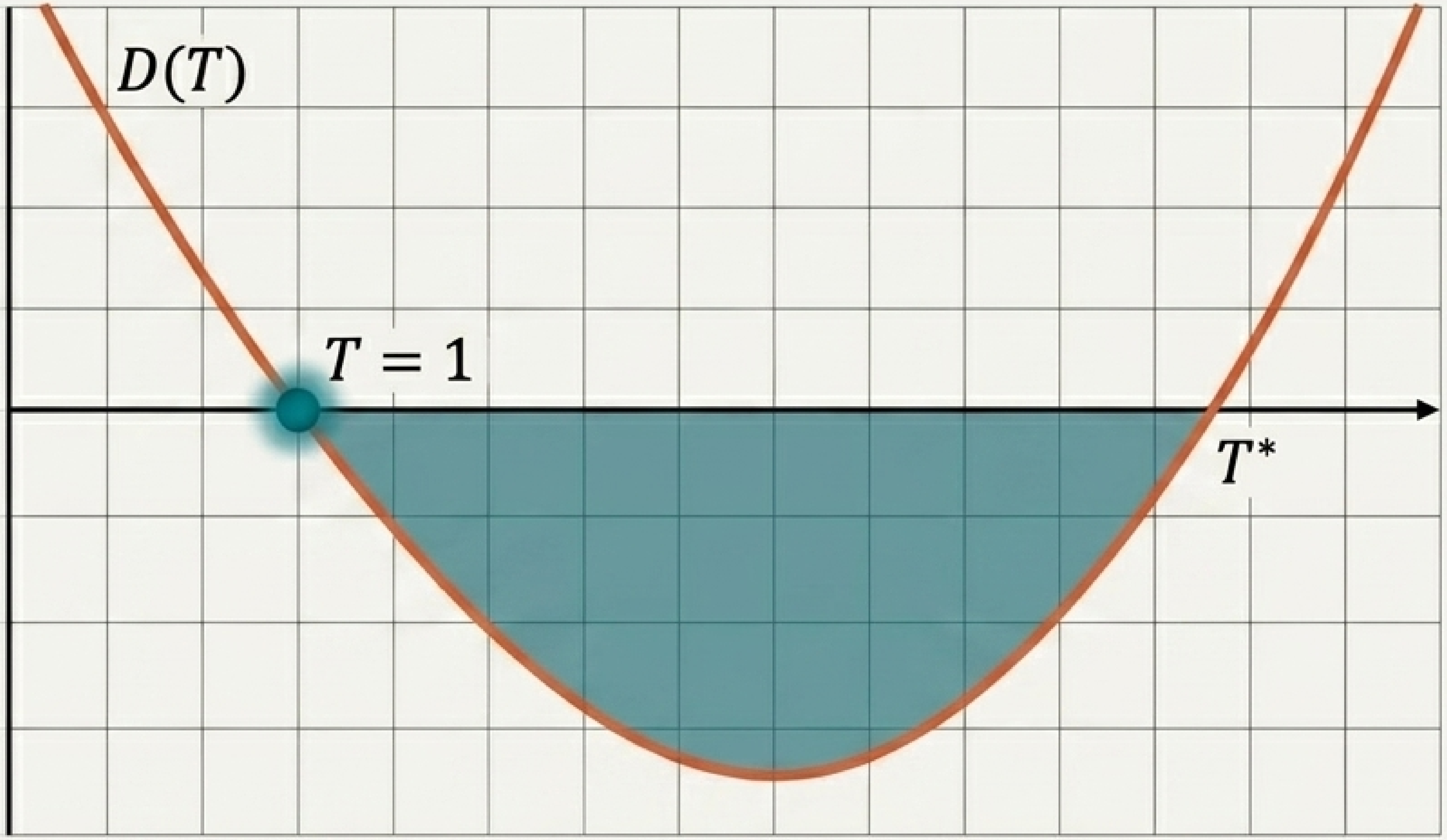


Regime 1: The Immediate Threshold.

The Mathematical Collapse:

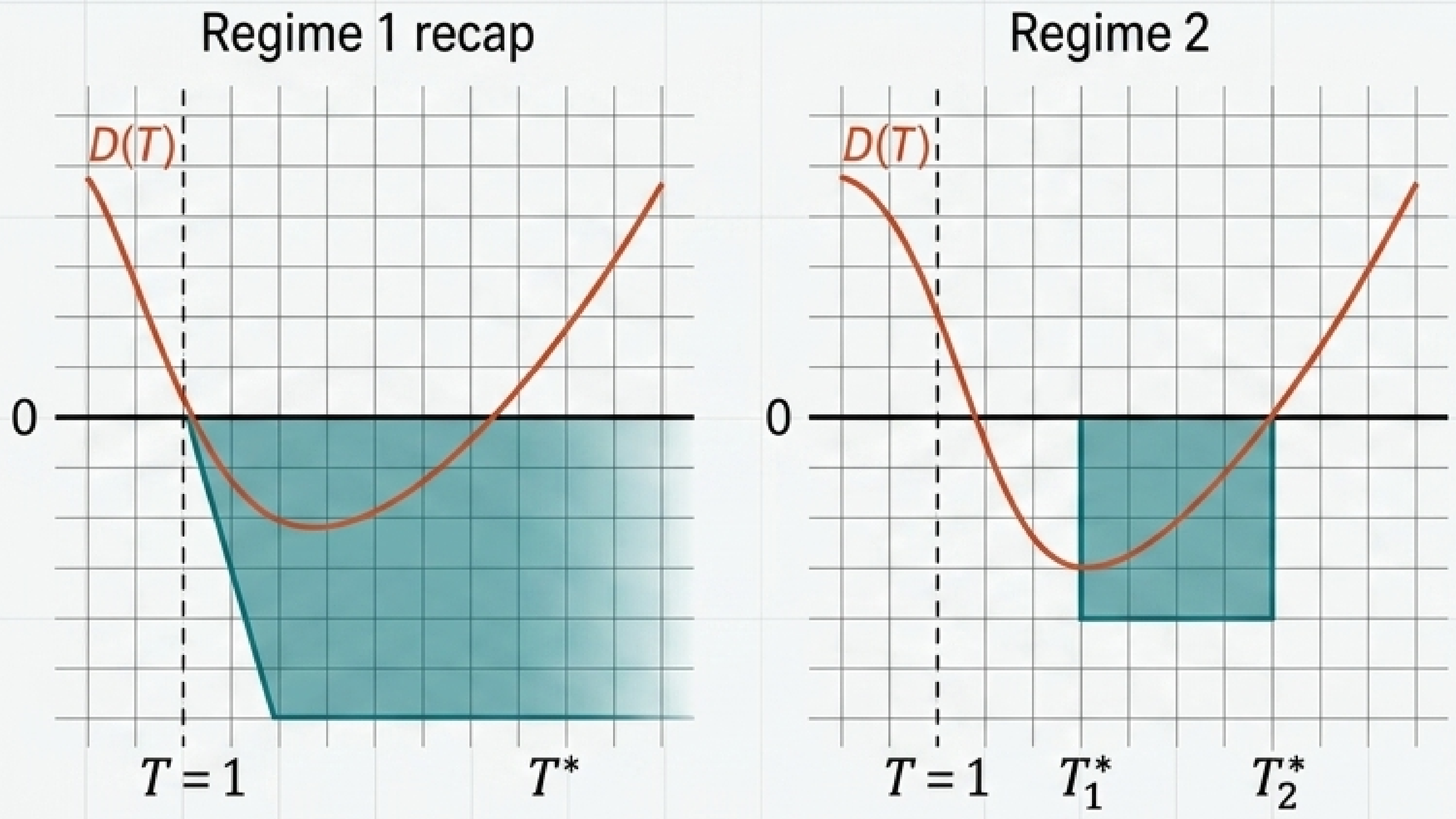
Evaluate D at $T = 1$. Because $\left((1+\delta)^1 - 1\right)^p = \delta^p$, dependencies on a, α, δ, p collapse perfectly.

$$D(1) = \kappa_R - \kappa_M$$



Core Insight: If repair overhead is cheaper than maintenance overhead ($\kappa_R < \kappa_M$), a deferral window opens immediately at $T = 1$. This is purely an overhead-driven phenomenon, completely independent of curvature or compounding.

Regime 2: The Late-Recovery Window.



Condition: $\kappa_R \geq \kappa_M$ (High repair overhead), but $T_{min} > 1$ (Early Recovery fails).

The Geometric Threshold: Instead of comparing κ_R to κ_M, we compare κ_R to κ_R^{crit}—a threshold derived from where the compounding curve's own minimum sits.

If $\kappa_R < \kappa_R^{crit}$, a deferral window opens later, but it is a finite interval (T_1^*, T_2^*). Deferral that becomes attractive late does not stay attractive indefinitely.

The Deferral Classification Matrix.

<table>
<tr><th></th><th>Column 1: $\kappa_R < \kappa_M$
(Low Repair Overhead)</th><th>Column 2: $\kappa_R \geq \kappa_M$
(High Repair Overhead)</th></tr>
<tr><td>Row 1: $x^* \leq \delta$
(Early Recovery)</td><td rowspan="2">Immediate Threshold
Favorable set: $[1, T^*)$
Permanent cutover.</td><td>Early-recovery, no threshold
Favorable set: ∅
Maintenance weakly dominates from start.</td></tr>
<tr><td>Row 2: $x^* > \delta$
(Late Recovery)</td><td>Late-recovery, window (if $\kappa_R < \kappa_R^{crit}$)
Favorable set: (T_1^*, T_2^*)
Finite interval.

Late-recovery, no window (if $\kappa_R \geq \kappa_R^{crit}$)
Favorable set: ∅</td></tr>
</table>

This classification is exhaustive and exact. The first threshold is governed by overheads; the second by geometry.

Asymptotic Anatomy of the Threshold.

$$T^* \approx T_1 - \frac{\kappa_R}{(p-1)(\kappa_M + \alpha a^p)} - \delta \frac{pT_1(T_1-1)}{2(p-1)}$$

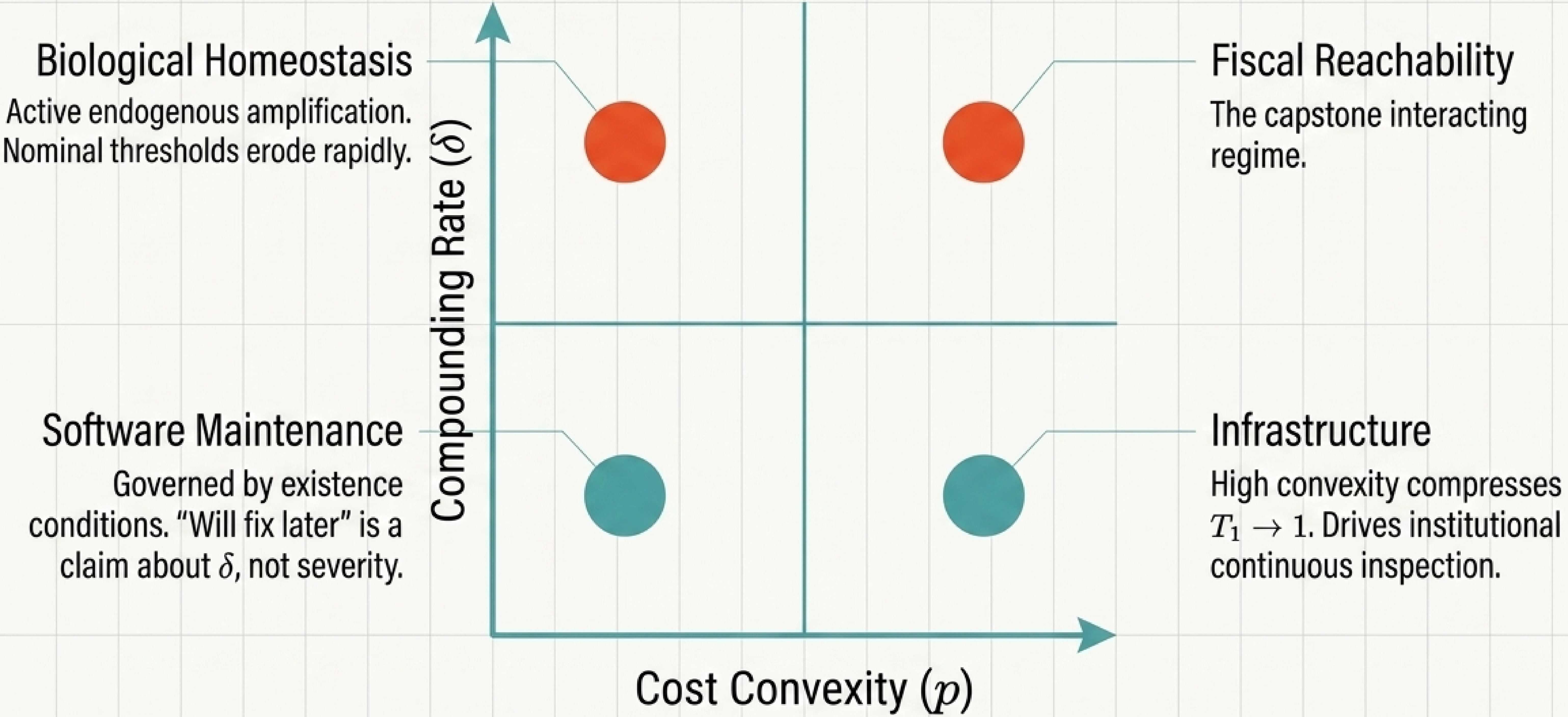
Baseline Deferral.
The pure-curvature tolerance
of the system when $\delta \to 0$.

Overhead Penalty.
Cheap-to-trigger repair
pulls the threshold down.

Compounding Penalty.
Note that it scales
with $T_1(T_1-1)$.

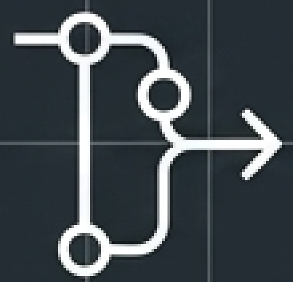
A system with a large baseline tolerance (T_1) loses its deferral window fastest under compounding.
Tolerance is eroded most rapidly in the systems that appear, on a static reading, most able to afford it.

The Parameter Landscape.



NotebookLM

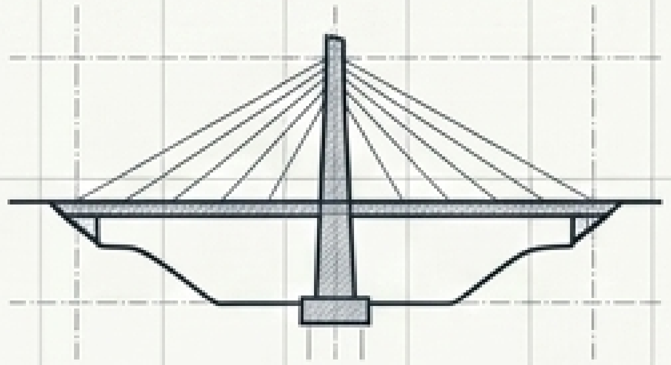
Parameter Extremes: Overhead vs. Convexity.



$\delta \approx 0$

Software

Governed entirely by overheads. If repair initiation (κ_R, shipping a hotfix) is cheaper than continuous monitoring (κ_M, CI overhead), batching minor defects is mathematically optimal.



Large p

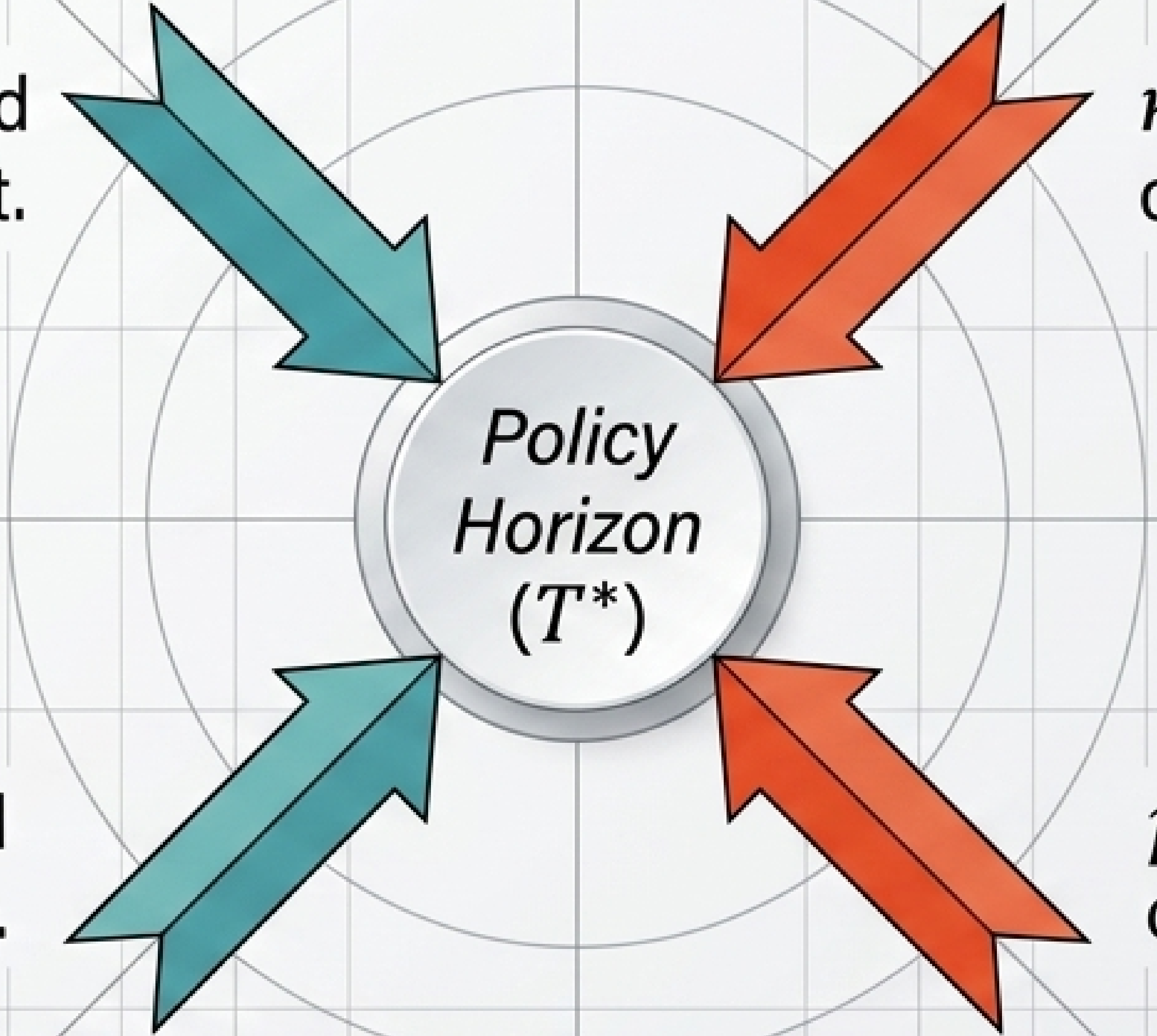
Infrastructure

Strongly convex repair costs disproportionately punish accumulated deviation. Large p compresses the baseline threshold T_1 toward 1, requiring continuous correction regardless of how slow individual defects grow.

The Capstone: Fiscal Reachability.

κ_M: Auditing and administrative oversight.

κ_R: Massive political initiation cost of emergency bailouts.



δ: Interest-on-arrears and eroding institutional credibility.

p: Strongly convex cost of crisis-stage correction.

In a fiscal system, T^* represents the **policy horizon**—the point where continuous stabilization becomes strictly cheaper than deferred intervention. Institutional credibility gives a false appearance of high tolerance (T_1) that **structural drift** (δ) rapidly destroys.

Redefining “Repair Delay”.

T

The Window of Deferral

Short of T, delay is not a defect.
It is the economically correct exercise of the deferral window.

The Failure Mode

Delay only becomes a systemic failure mode once the computable threshold T has been crossed.

Rational system design does not eliminate delay; it precisely calculates its mathematical boundary.